2022012233 芦锦森 HW3 数值分析

exercise-1s2

1. $\det A = 50 - ac \neq 0 \Rightarrow ac \neq 50$.

(1) Jacobi 迭代法: $B_J = \begin{pmatrix} 10 & \\ & 5 \end{pmatrix}^{-1}\begin{pmatrix} & -a \\ -c & \end{pmatrix} = \begin{pmatrix} & -\frac{1}{10}a \\ -\frac{1}{5}c & \end{pmatrix}$.

$C_{B_J}(\lambda) = \det(\lambda I - B_J) = \det\begin{pmatrix} \lambda & \frac{a}{10} \\ \frac{c}{5} & \lambda \end{pmatrix} = \lambda^2 - \frac{ac}{50} = 0 \Rightarrow |\lambda| = \sqrt{\frac{|ac|}{50}}$.

$\Rightarrow \rho(B_J) = \sqrt{\frac{|ac|}{50}} < 1 \Rightarrow |ac| < 50$.

(2) Gauss-Seidel 迭代法: $B_{GS} = \begin{pmatrix} 10 & \\ c & 5 \end{pmatrix}^{-1}\begin{pmatrix} & -a \\ & \end{pmatrix} = \begin{pmatrix} 0 & -\frac{a}{10} \\ 0 & \frac{ac}{50} \end{pmatrix}$.

$C_{B_{GS}}(\lambda) = \det(\lambda I - B_{GS}) = \det\begin{pmatrix} \lambda & \frac{a}{10} \\ 0 & \lambda - \frac{ac}{50} \end{pmatrix} = \lambda(\lambda - \frac{ac}{50}) = 0 \Rightarrow \lambda = 0$ 或 $\frac{ac}{50}$.

$\Rightarrow \rho(B_{GS}) = \frac{|ac|}{50} < 1 \Rightarrow |ac| < 50$.

2. $A = D - L - U = D(I - \tilde{L} - \tilde{U})$.

对称正定. $\begin{cases} L = D\tilde{L} \\ U = D\tilde{U} \end{cases}$

$$\begin{cases} (D-L)x^{(k+\frac{1}{2})} = Ux^{(k)} + b \\ (D-U)x^{(k+1)} = Lx^{(k+\frac{1}{2})} + b. \end{cases}$$

(1). $x^{(k+1)} = (D-U)^{-1}Lx^{(k+\frac{1}{2})} + (D-U)^{-1}b$

$= (D-U)^{-1}L(D-L)^{-1}Ux^{(k)} + (D-U)^{-1}L(D-L)^{-1}b + (D-U)^{-1}b$

$= (I-\tilde{U})^{-1}D^{-1}D\tilde{L}(I-\tilde{L})^{-1}D^{-1}D\tilde{U}x^{(k)} + (I-\tilde{U})^{-1}D^{-1}D(I-\tilde{L})^{-1}D^{-1}b$

$= (I-\tilde{U})^{-1}\tilde{L}(I-\tilde{L})^{-1}\tilde{U}x^{(k)} + (I-\tilde{U})^{-1}(I-\tilde{L})^{-1}D^{-1}b$.

由 $(I-\tilde{L})\tilde{L} = \tilde{L} - \tilde{L}^2 = \tilde{L}(I-\tilde{L}) \Rightarrow \tilde{L}(I-\tilde{L})^{-1} = (I-\tilde{L})^{-1}\tilde{L}$

$= \underbrace{(I-\tilde{U})^{-1}(I-\tilde{L})^{-1}\tilde{L}\tilde{U}}_{B}x^{(k)} + \underbrace{(I-\tilde{U})^{-1}(I-\tilde{L})^{-1}D^{-1}b}_{g}$

$\triangleq Bx^{(k)} + g$.



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(2). 设 $Bx=\lambda x$，其中 $x\in\mathbb{C}^n$，$\lambda\in\mathbb{C}$，$x\neq 0$.

$\Rightarrow (I-\tilde{U})^{-1}(I-\tilde{L})^{-1}\tilde{L}\tilde{U}x=\lambda x$

$\Rightarrow \tilde{L}\tilde{U}x=\lambda(I-\tilde{L})(I-\tilde{U})x=\lambda\left(I-\tilde{L}-\tilde{U}+\tilde{L}\tilde{U}\right)x$

$\Rightarrow (1-\lambda)\tilde{L}\tilde{U}x=\lambda(I-\tilde{L}-\tilde{U})x$

$\Rightarrow (1-\lambda)D^{-1}LD^{-1}Ux=\lambda D^{-1}Ax$

$\Rightarrow (1-\lambda)LD^{-1}Ux=\lambda Ax$.

$\Rightarrow (1-\lambda)\bar{x}^TLD^{-1}Ux=\lambda\bar{x}^TAx$. (*)

由于 A 为实对称正定矩阵，故 A 也复正定. (具体：作正交对角化 $Q\in\mathbb{R}^{n\times n}$ 正交，$QAQ^T=\Lambda=(\lambda_1\cdots\lambda_n)$

$\Rightarrow \bar{x}^TAx>0$.

$\Rightarrow \bar{x}^TAx=\bar{x}^TQ^TAQx=\overline{(Qx)}^TA(Qx)$

$\triangleq \bar{y}^T\Lambda y=\sum_{i=1}^n\lambda_i|y_i|^2>0$.)

另一方面，由于 $L=U^T=\bar{U}^T$ $(U\in\mathbb{R}^{n\times n})$，$D^{-1}$ 对称正定

$\Rightarrow \bar{x}^TLD^{-1}Ux=(\overline{Ux})^TD^{-1}(Ux)\geq 0$.

若 $\lambda=1$，由(*) $\Rightarrow \lambda=0$，矛盾. 故由(*) $\Rightarrow \frac{\lambda}{1-\lambda}=\frac{\bar{x}^TLD^{-1}Ux}{\bar{x}^TAx}\geq 0$

设 $\frac{\lambda}{1-\lambda}=a\in\mathbb{R}_{\geq 0}$ $\Rightarrow \lambda=a-a\lambda\Rightarrow\lambda=\frac{a}{1+a}\geq 0$. 为非负实数.

↳ $a=a(\lambda)$ 与 $\lambda$ 有关.

(3). 由(2)知，$\forall B$ 特征值，$\exists a\geq 0$ s.t. $\lambda=\frac{a}{1+a}<1$

$\Rightarrow \rho(B)<1$，故迭代收敛.

3. SOR 迭代法：$B_{SOR}^{\omega}=(D-\omega L)^{-1}[(1-\omega)D+\omega U]$.

A 严格对角占优，$0<\omega\leq 1$.

设 $B_{SOR}^{\omega}x=\lambda x$，$\lambda\in\mathbb{C}$，$0\neq x\in\mathbb{C}^n$.

$\Rightarrow [(1-\omega)D+\omega U]x=\lambda(D-\omega L)x$

$\Rightarrow [(1-\omega-\lambda)D+\omega U+\omega\lambda L]x=0$.



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反证，若 $|\lambda|\geq 1$，则 $1-\omega-\lambda\neq 0$.

$$\Rightarrow \left(D+\frac{\omega U+\omega\lambda L}{1-\omega-\lambda}\right)x=0. \quad (*)$$

由于 $\left|\frac{\omega}{1-\omega-\lambda}\right|\leq 1 \iff \omega\leq|1-\omega-\lambda| \iff \omega\leq \operatorname{dist}(1-\omega, \partial B(0,1)) = 1-(1-\omega)=\omega$ 成立，（$1-\omega\in[0,1)$）

且 $\left|\frac{\omega\lambda}{1-\omega-\lambda}\right|\leq 1 \iff \omega^2|\lambda|^2\leq(1-\omega-\lambda)(1-\omega-\bar{\lambda}) = (1-\omega)^2-2\mathrm{Re}(\lambda)(1-\omega)+|\lambda|^2$

$$\iff (|\lambda|^2-1)\omega^2+(2-2\mathrm{Re}(\lambda))\omega-1+2\mathrm{Re}(\lambda)-|\lambda|^2\leq 0.$$

此为关于 $\omega$ 的一次函数或开口向上的二次函数，故只需验证 $0<\omega\leq 1$ 两端点处成立即可：

- $\omega=0 \iff -1+2\mathrm{Re}\omega-|\lambda|^2\leq 0 \iff (\lambda-1)(\bar{\lambda}-1)\geq 0 \iff |\lambda-1|^2\geq 0$ 成立.
- $\omega=1 \iff |\lambda|^2-1+2-2\mathrm{Re}(\lambda)-1+2\mathrm{Re}(\lambda)-|\lambda|^2\leq 0 \iff 0\leq 0$ 成立.

从而对矩阵 $B\triangleq D+\frac{\omega U+\omega\lambda L}{1-\omega-\lambda}$，

$$\sum_{i\neq k}|b_{ki}| \leq \left|\frac{\omega\lambda}{1-\omega-\lambda}\right|\cdot\sum_{1\leq i\leq k-1}|a_{ki}| + \left|\frac{\omega}{1-\omega-\lambda}\right|\sum_{k+1\leq i\leq n}|a_{ki}| \leq \sum_{i\neq k}|a_{ki}| < |a_{kk}| = |b_{kk}|$$

（由A严格对角占优）

$\Rightarrow$ B 严格对角占优，从而可逆.

从而由 $(*)$：$Bx=0 \Rightarrow x=0$，矛盾！

故 $|\lambda|<1$，从而 $\rho(B_{SOR}^{\omega})<1$，迭代收敛.

4. A 实对称正定. 设 A 特征值 $0<\lambda_n\leq\lambda_{n-1}\leq\cdots\leq\lambda_1=\rho(A)$.

$$x^{(k+1)}=x^{(k)}+\omega(b-Ax^{(k)}) = (I-\omega A)x^{(k)}+\omega b.$$

$B=I-\omega A$. 若 $\omega=0$ 显然不收敛.

设 $Bx=\lambda x \Rightarrow x-\omega Ax=\lambda x \Rightarrow Ax=\frac{1-\lambda}{\omega}x$.

故 $\frac{1-\lambda}{\omega}$ 是 A 特征值. $\frac{1-\lambda}{\omega}=\lambda_i$ $(1\leq i\leq n)$. $\Rightarrow \lambda=1-\omega\lambda_i$.

要使迭代收敛，则 $|\lambda|<1$，即 $|1-\omega\lambda_i|<1$，$1\leq i\leq n$.

则 $\omega>0$；再结合函数图像，只需满足 $|1-\omega\lambda_1|<1 \iff \omega<\frac{2}{\lambda_1}=\frac{2}{\rho(A)}$.

即迭代收敛 $\iff 0<\omega<\frac{2}{\rho(A)}$.



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5. A对称正定. 共轭梯度法: $\alpha_k = \frac{(r^{(k)}, p^{(k)})}{(Ap^{(k)}, p^{(k)})}$, $x^{(k+1)} = x^{(k)} + \alpha_k p^{(k)}$.

$$\begin{aligned} J(x^{(k+1)}) - J(x^{(k)}) &= \frac{1}{2}(Ax^{(k)} + \alpha_k Ap^{(k)}, x^{(k)} + \alpha_k p^{(k)}) - \frac{1}{2}(Ax^{(k)}, x^{(k)}) \\ &\quad - (b, x^{(k)} + \alpha_k p^{(k)}) + (b, x^{(k)}) \\ &= \frac{1}{2}\alpha_k^2 (Ap^{(k)}, p^{(k)}) + \alpha_k (Ax^{(k)}, p^{(k)}) - \alpha_k (b, p^{(k)}) \\ &= \frac{\alpha_k^2}{2}(Ap^{(k)}, p^{(k)}) - \alpha_k (r^{(k)}, p^{(k)}) \\ &= \frac{\alpha_k^2}{2}(Ap^{(k)}, p^{(k)}) - \alpha_k^2 (Ap^{(k)}, p^{(k)}) \\ &= -\frac{\alpha_k^2}{2}(Ap^{(k)}, p^{(k)}) \le 0 \quad (\text{由 } A \text{ 正定}). \end{aligned}$$

取等 $\Leftrightarrow \alpha_k p^{(k)} = 0$ $\begin{cases} \alpha_k = 0 \Rightarrow (r^{(k)}, p^{(k)}) = 0 \\ p^{(k)} = 0 \Rightarrow (r^{(k)}, p^{(k)}) = 0 \end{cases}$

$$\begin{aligned} \Rightarrow 0 = (r^{(k)}, p^{(k)}) &= (r^{(k)}, r^{(k)} + \beta_{k-1} p^{(k-1)}) \\ &= (r^{(k)}, r^{(k)}) + \beta_{k-1}(r^{(k)}, p^{(k-1)}) \\ &= (r^{(k)}, r^{(k)}) \end{aligned}$$

由共轭梯度法过程 $p^{(0)} = r^{(0)}$, $p^{(k+1)} = r^{(k+1)} + \beta_k p^{(k)}$

$\Rightarrow p^{(k-1)}$ 是 $r^{(0)}, \dots, r^{(k-1)}$ 线性组合.

又 $\{r^{(k)}\}$ 正交 $\Rightarrow (r^{(k)}, p^{(k-1)}) = 0$.

$\Rightarrow r^{(k)} = 0$.

从而若 $r^{(k)} \neq 0$, 则为严格不等号.

6. Galerkin 原理: 在 K 中找近似解 z s.t. $r - Az \perp L$. A对称正定.

(1) 由 $p \in K$, 设 $p = sr + tAr$.

$0 = (r, Ap) = s(r, Ar) + t(r, A^2 r)$.

取 $s = 1$, 则 $t = -\frac{(r, Ar)}{(r, A^2 r)} = -\frac{(r, Ar)}{(Ar, Ar)}$.

$\Rightarrow p = r - \frac{(r, Ar)}{(Ar, Ar)} Ar$.

(2). 设近似解 $z = sr + tp$.

则由 $r - Az \perp L$ $\Rightarrow$ $\begin{cases} (r - sAr - tAp, r) = 0 \\ (r - sAr - tAp, p) = 0 \end{cases}$



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$$\Rightarrow \begin{cases} (r,Ar)s + (r,Ap)t = (r,r) \\ 0 = (p,Ar)s + (p,Ap)t = (r,p) \end{cases}$$

$$\Rightarrow \begin{cases} s = \dfrac{(r,r)}{(r,Ar)} \\ t = \dfrac{(r,p)}{(p,Ap)} \end{cases}.$$

从而 $x_1 = x_0 + z = x_0 + \dfrac{(r,r)}{(r,Ar)} r + \dfrac{(r,p)}{(p,Ap)} p$，其中 $p$ 由 (1) 给出.

(3) 迭代式：$x_{n+1} = x_n + \dfrac{(r_n, r_n)}{(r_n, Ar_n)} r_n + \dfrac{(r_n, p_n)}{(p_n, Ap_n)} p_n$,

其中 $r_n = b - Ax_n$,

$$p_n = r_n - \frac{(r_n, Ar_n)}{(Ar_n, Ar_n)} Ar_n.$$

由前述 Galerkin 原理过程，有 $(r_n, Ap_n) = 0$，$r_{n+1} \perp \mathrm{span}\{r_n, p_n\}$

即 $(r_{n+1}, r_n) = 0$，$(r_{n+1}, p_n) = 0$.

$$r_{n+1} = r_n - \frac{(r_n, r_n)}{(r_n, Ar_n)} Ar_n - \frac{(r_n, p_n)}{(p_n, Ap_n)} Ap_n.$$

考虑 $(r_{n+1}, A^{-1}r_{n+1}) = (r_{n+1}, A^{-1}r_n) - \underbrace{\left(r_{n+1}, \dfrac{(r_n,r_n)}{(r_n,Ar_n)} r_n + \dfrac{(r_n,p_n)}{(p_n,Ap_n)} p_n\right)}_{=0}$

$$= (r_{n+1}, A^{-1}r_n)$$

$$= (r_n, A^{-1}r_n) - \frac{(r_n,r_n)}{(r_n,Ar_n)} \underbrace{(Ar_n, A^{-1}r_n)}_{=(r_n,r_n)} - \frac{(r_n,p_n)}{(p_n,Ap_n)} \underbrace{(Ap_n, A^{-1}r_n)}_{=(p_n,r_n)=(r_n,p_n)}$$

$$= (r_n, A^{-1}r_n) - \frac{(r_n,r_n)^2}{(r_n,Ar_n)} - \frac{(r_n,p_n)^2}{(p_n,Ap_n)}$$

$$\leq (r_n, A^{-1}r_n).$$

即数列 $\{(r_n, A^{-1}r_n)\}_n$ 递减，且由于 $A^{-1}$ 正定故有下界 0.

$\Rightarrow \{(r_n, A^{-1}r_n)\}_n$ 收敛，从而同时有 $\lim\limits_{n\to\infty} \dfrac{(r_n,r_n)^2}{(r_n,Ar_n)} = 0$.



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由 A 对称正定，$\rho(A)>0$，A 的 n 个互异 n 个特征值 ($\in \mathbb{R}_{>0}$) 的特征向量组成 $\mathbb{R}^n$ 正交基.

↳ $(\lambda_i, x_i)$，$\|x_i\|=1$ 单位

从而可写 $r_n = \sum_{i=1}^{n} a_i x_i$，其中 $a_i \in \mathbb{R}$

$$\Rightarrow (r_n, Ar_n) = \left(\sum_{i=1}^{n} a_i x_i, \sum_{i=1}^{n} a_i A x_i\right) = \left(\sum_{i=1}^{n} a_i x_i, \sum_{i=1}^{n} a_i \lambda_i x_i\right)$$

$$= \sum_{i=1}^{n} (a_i x_i, a_i \lambda_i x_i) = \sum_{i=1}^{n} \lambda_i (a_i x_i, a_i x_i) \leq \sum_{i=1}^{n} \rho(A)\,(a_i x_i, a_i x_i)$$

$$= \rho(A)\cdot \left(\sum_{i=1}^{n} a_i x_i, \sum_{i=1}^{n} a_i x_i\right) = \rho(A)\cdot (r_n, r_n).$$

$$\Rightarrow (r_n, r_n) \leq \rho(A)\cdot \frac{(r_n, r_n)^2}{(r_n, Ar_n)} \to 0 \quad (n\to\infty).$$

$\Rightarrow r_n \to 0\ (n\to\infty)$，从而 $Ax_n \to b\ (n\to\infty)$，即该算法收敛.

7. (选做). A 对称正定. 设所有特征值为 $\rho(A)=\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n > 0$.

$x^{(k+1)} = x^{(k)} - \alpha r^{(k)} + \beta(x^{(k)} - x^{(k-1)})$. 其中 $r^{(k)} = b - Ax^{(k)}$.

$$\Rightarrow r^{(k+1)} = r^{(k)} + \alpha A r^{(k)} + \beta(r^{(k)} - r^{(k-1)}).$$

$$\Rightarrow r^{(k+1)} = \left((1+\beta)I + \alpha A\right) r^{(k)} - \beta r^{(k-1)} \quad (*)$$

将 A 正交对角化：$Q\cdot AQ^T = \Lambda = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$.

$$\Rightarrow r^{(k+1)} = \left[(1+\beta)I + Q^T \Lambda Q\right] r^{(k)} - \beta r^{(k-1)}.$$

$$\Rightarrow \underline{Q r^{(k+1)}} = \left[(1+\beta)I + \Lambda\right] \underline{Q r^{(k)}} - \beta \underline{Q r^{(k-1)}}$$

由于 Q 正交从而保 2-范数，从而考虑 $\{r^{(k)}\}$ 的收敛性与 $\{Qr^{(k)}\}$ 等价.

因此可不妨直接假设 $A = \Lambda = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$，

则 $(*) \Leftrightarrow r_i^{(k+1)} = (1+\beta+\alpha\lambda_i) r_i^{(k)} - \beta r_i^{(k-1)}$，$1\leq i\leq n$.

$$\Leftrightarrow \begin{pmatrix} r_i^{(k+1)} \\ r_i^{(k)} \end{pmatrix} = \begin{pmatrix} 1+\beta+\alpha\lambda_i & -\beta \\ 1 & 0 \end{pmatrix} \begin{pmatrix} r_i^{(k)} \\ r_i^{(k-1)} \end{pmatrix}.$$

记 $B_i = \begin{pmatrix} 1+\beta+\alpha\lambda_i & -\beta \\ 1 & 0 \end{pmatrix}$，该迭代收敛 $\Leftrightarrow \rho(B_i) < 1$，$\forall 1\leq i\leq n$.



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$C_{B_i}(\lambda) = \det(\lambda I - B_i) = \left(\lambda - (1+\beta+\alpha\lambda_i)\right)\lambda + \beta = \lambda^2 - (1+\alpha\lambda_i+\beta)\lambda + \beta.$

$\triangleq \lambda^2 - \gamma_i\lambda + \beta.$

记 $\gamma_i = 1+\alpha\lambda_i+\beta \in \mathbb{R}$. 则 $\lambda_{B_i} = \dfrac{\gamma_i \pm \sqrt{\gamma_i^2 - 4\beta}}{2}$.

① 当 $\gamma_i^2 - 4\beta \geq 0$ 时，两实根 $\Rightarrow \left|\dfrac{\gamma_i \pm \sqrt{\gamma_i^2 - 4\beta}}{2}\right| < 1$

$\Rightarrow \max\{-2-\gamma_i, -2+\gamma_i\} < \sqrt{\gamma_i^2 - 4\beta} < \min\{2-\gamma_i, 2+\gamma_i\}$

i) $\gamma_i \geq 0 \Rightarrow 0 \leq \gamma_i < 2$，且 $\gamma_i < \beta+1 \Rightarrow 0 \leq \gamma_i < \min\{2, 1+\beta\}$

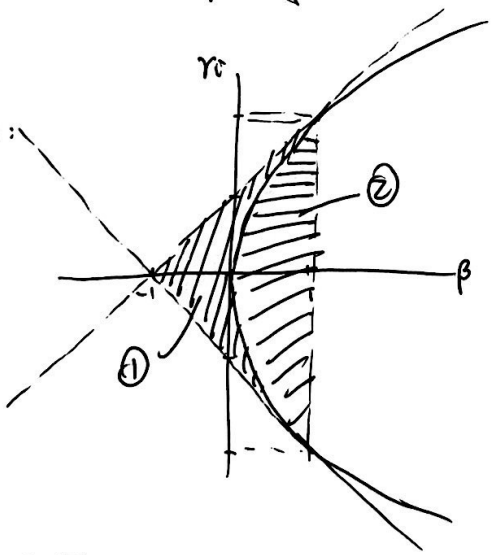
ii) $\gamma_i \leq 0 \Rightarrow \max\{-2, -1-\beta\} < \gamma_i \leq 0$.

从而此情形 $\lambda_i$ 取值范围 $\begin{cases} \gamma_i^2 \geq 4\beta \\ \max\{-2, -1-\beta\} < \gamma_i < \min\{2, 1+\beta\}. \end{cases}$

② 当 $\gamma_i^2 - 4\beta \leq 0$ 时，两虚根，$|\lambda_{B_i}| = \sqrt{\beta} < 1 \Rightarrow 0 \leq \beta < 1$. （$\beta \geq 0$）

从而此情形 $\lambda_i$ 取值范围 $\begin{cases} \gamma_i^2 \leq 4\beta \\ 0 \leq \beta < 1. \end{cases}$

作出 $\beta - \gamma_i$ 图：




故 ① + ② 取值范围为 $\begin{cases} -1 < \beta < 1 \\ |\gamma_i| < 1+\beta \end{cases} \Rightarrow |1+\beta+\alpha\lambda_i| < 1+\beta$

$\Rightarrow \alpha < 0$，且 $\alpha > -\dfrac{2(1+\beta)}{\lambda_i}$.

上式对 $1 \leq i \leq n$ 均成立，从而最终取值范围为

$$\begin{cases} -1 < \beta < 1 \\ -\dfrac{2(1+\beta)}{\rho(A)} < \alpha < 0 \end{cases}$$

最终 $\alpha, \beta$ 取值不依赖 $\lambda_i$.



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